# ECE 340: Semiconductor Devices

## Lecture 6: Intrinsic and Extrinsic Materials

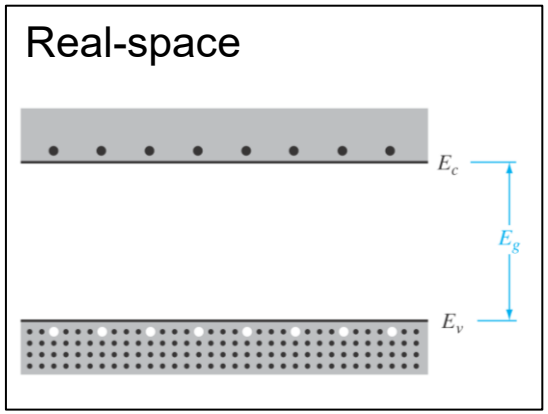
**Prof. Hyunseok Kim**

**Lecture (Section C):** M,W,F / 2:00-2:50pm / 3017 ECEB

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# (Recap) E-k diagram and Effective Mass




k is the wavenumber

$$k = \frac{2\pi}{\lambda} = \frac{p}{\hbar}$$

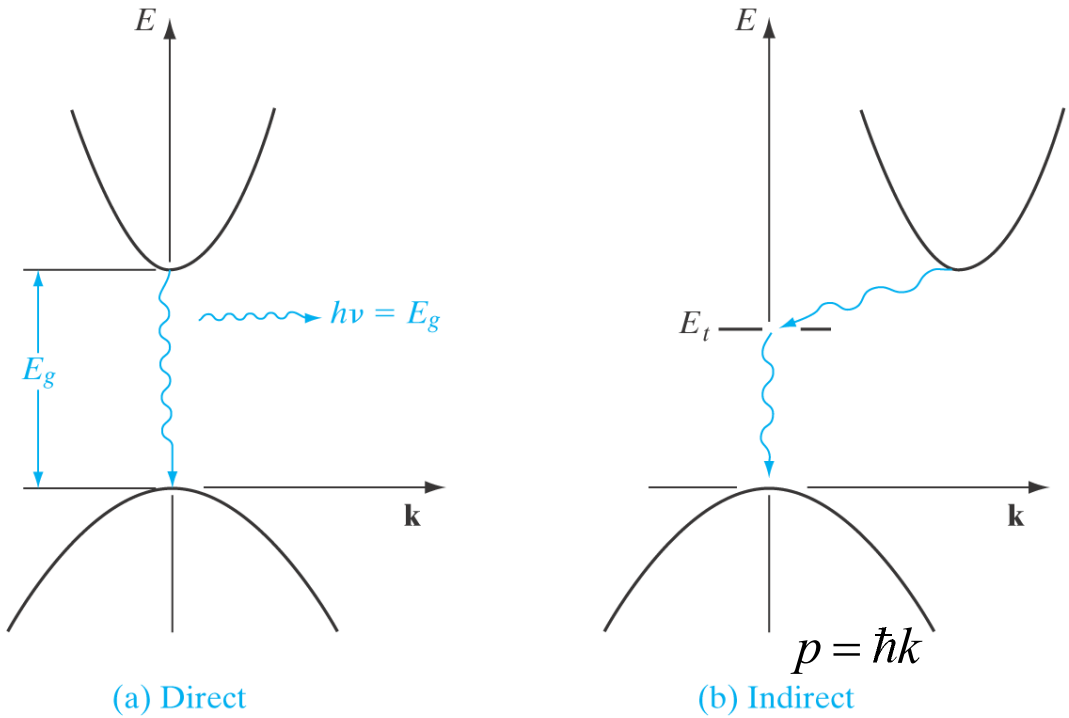



Figure 3.5

Direct and indirect electron transitions in semiconductors: (a) direct transition with accompanying photon emission; (b) indirect transition via a defect level.

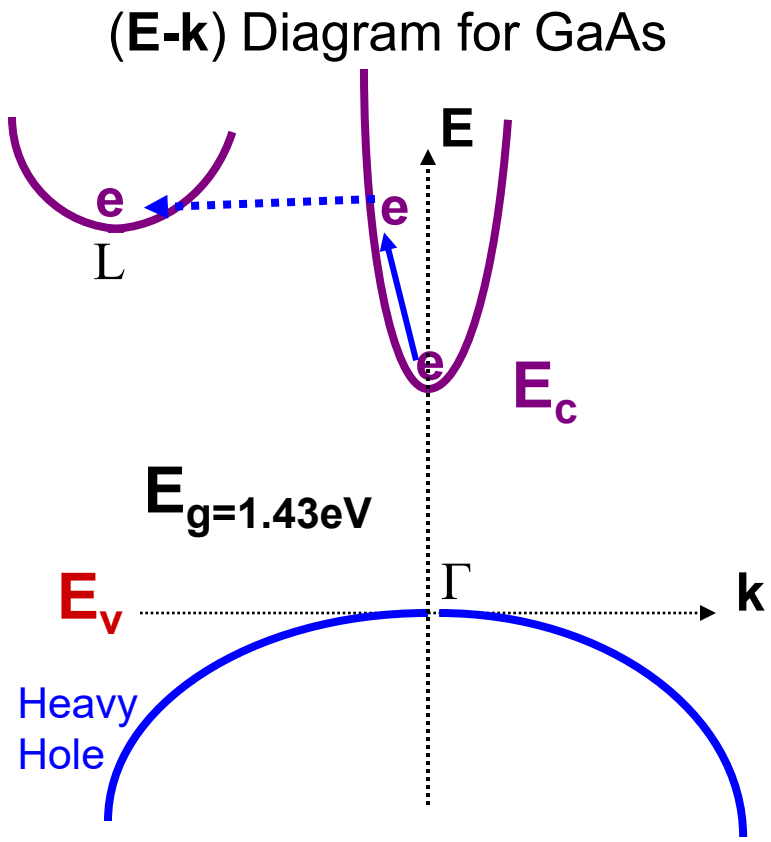
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# (Recap) E-k diagram and Effective Mass

- **The curvature of the parabola differs from the free space curvature**, and depends upon which minima the electron or hole is in
- Near a minima, the E(k) relationship is:

$$E = \frac{\hbar^2}{2m^*}\mathbf{k}^2$$

- This is similar to the E(k) relationship for a free electron with a mass m* instead of $m_o$
- The parameter **m*** is the **effective mass**
- What about valence band? → **Positive "Hole" effective mass**

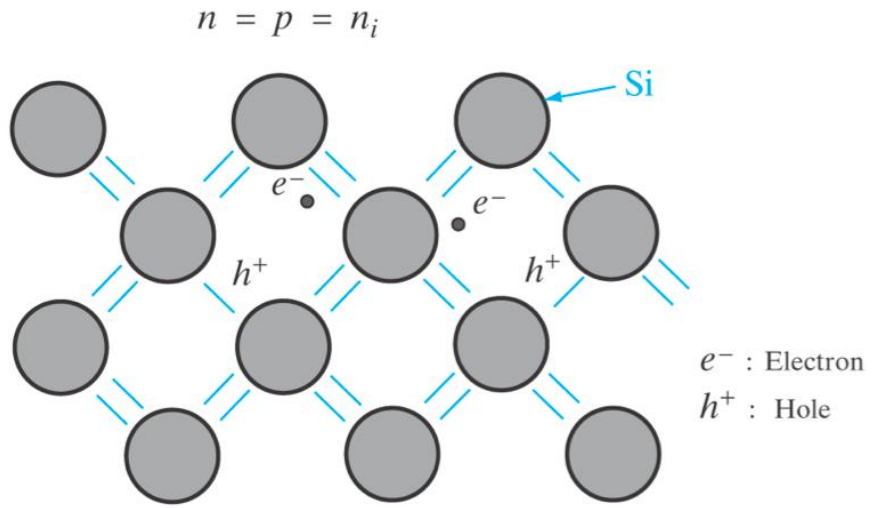


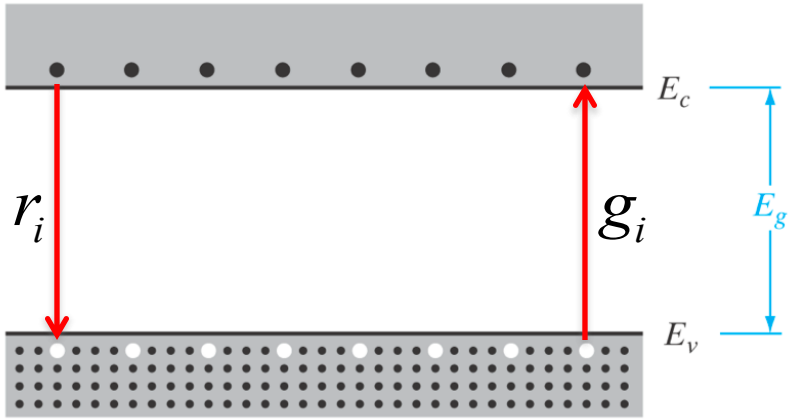

| | Ge | Si | GaAs |
|---|---|---|---|
| $m_n^*$(or $m_e^*$) | $0.55m_o$ | $1.1m_o$ | $0.067m_o$ |
| $m_p^*$(or $m_h^*$) | $0.37m_o$ | $0.56m_o$ | $0.48m_o$ |

# Intrinsic and Extrinsic Material

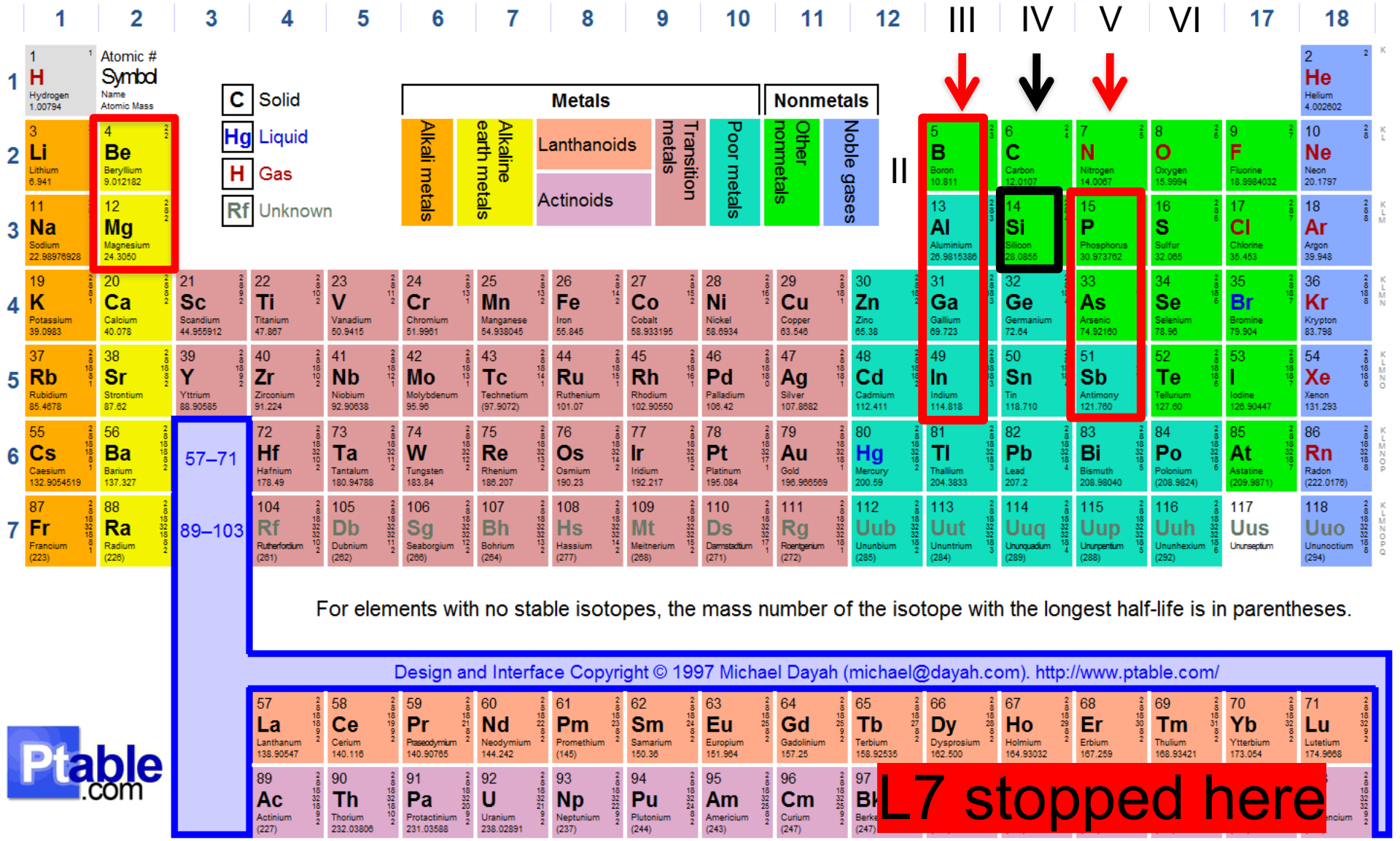
# Intrinsic Material

- Intrinsic material: No impurity (no donor or acceptor)
- Electrons and holes present in equal concentrations, called the ***intrinsic concentration, $n_i$***
- $n_i$ of Si at 300K ~ $1.5 \times 10^{10}$ $cm^{-3}$
- The intrinsic concentration $n_i$ increases with increasing temperature (more thermal energy)
- The intrinsic calculation is based upon thermal energy only – no light, current injection, etc.



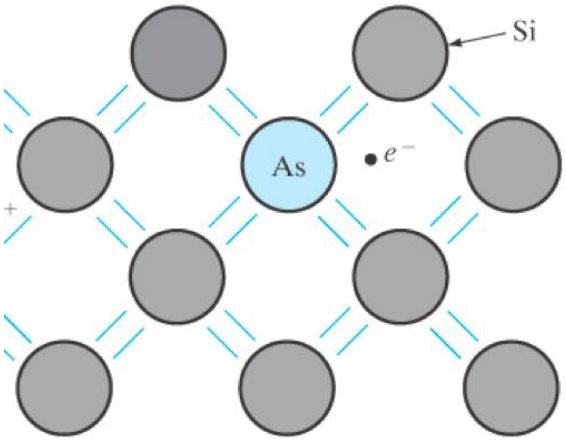



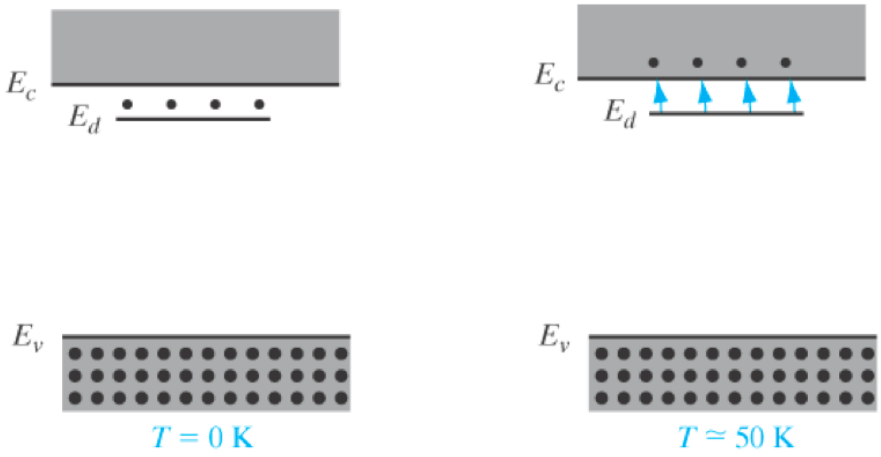
# What Happens if We Insert an Atom from Another Column?



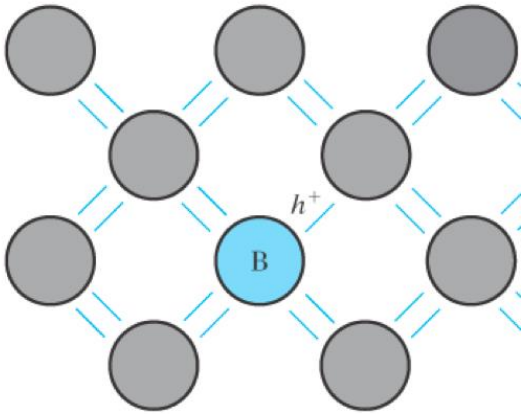
# Extrinsic Material

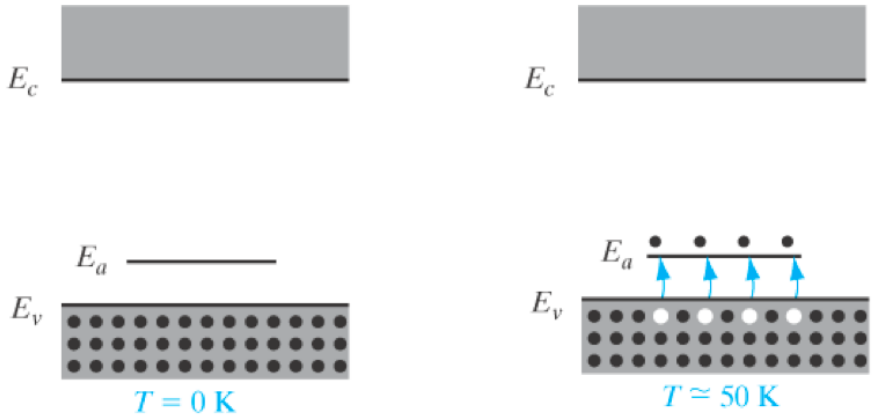
As impurity (donor)




B impurity (acceptor)



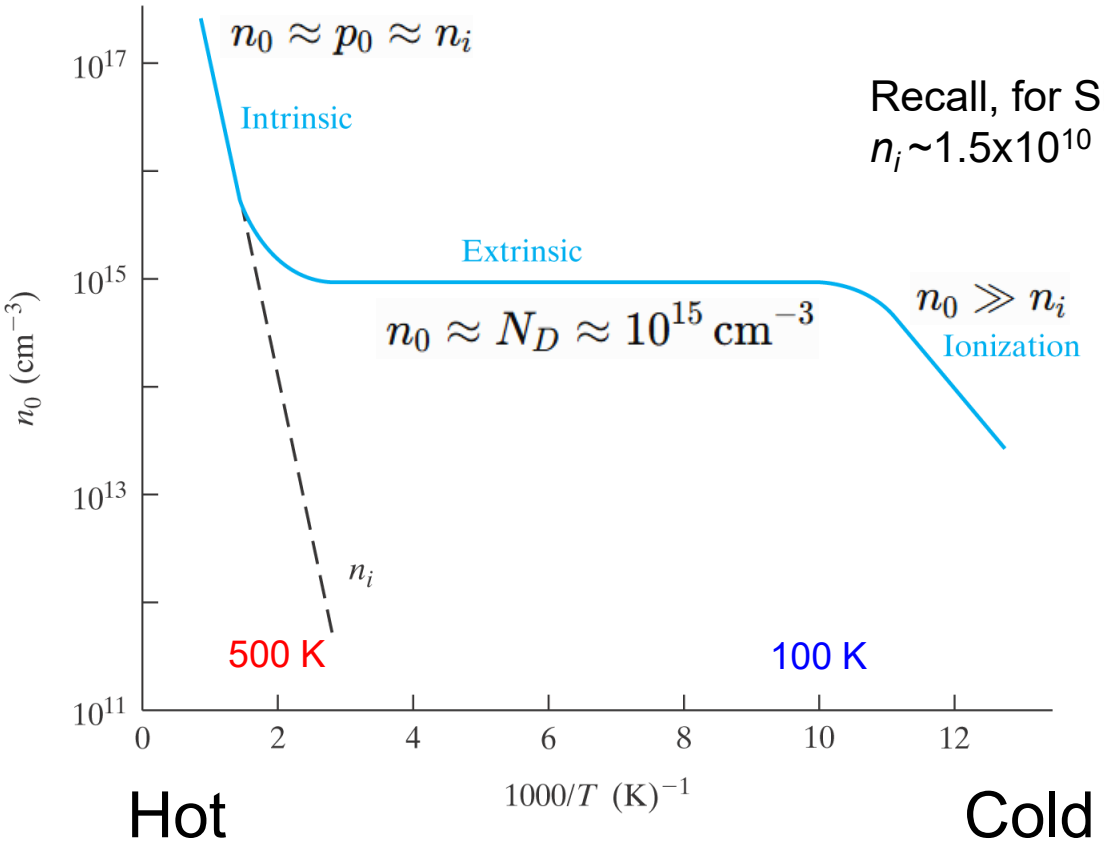




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# Extrinsic Material: Temperature Effects

Carrier concentration in Si doped with $10^{15}$ donors/cm$^3$

$n_0$: Electron concentration (in conduction band)




Recall, for Si, $n_i$ ~1.5x10$^{10}$ cm$^{-3}$ at 300K

Figure 3.18

Carrier concentration vs. inverse temperature for Si doped with $10^{15}$ donors/cm$^3$.

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# Important Definitions

- **Intrinsic Material:** For a "perfect" crystal in equilibrium, electron and hole densities are equal
  - The value of this density $n_i$ is called the intrinsic concentration
- **Extrinsic Material:** The crystal is "doped" with
  - **donors**, where the material is n-type and the electron density exceeds the hole density, or
  - **acceptors**, where the material is p-type and the hole density exceeds the electron density
  - An **amphoteric dopant** can act as either a donor or acceptor, depending upon what atom it is replacing in the crystal
  - In extrinsic material, the carrier with the higher concentration is the **majority carrier** and the carrier with lower concentration is the **minority carrier**
    - We will find out later that the product of the minority carrier density and majority carrier density is constant for a given temperature (in equilibrium), and equal to $n_i^2$
    - In p-type material, holes are the majority carrier, electrons the minority carrier
    - In n-type material, electrons are the majority carrier, holes the minority carrier

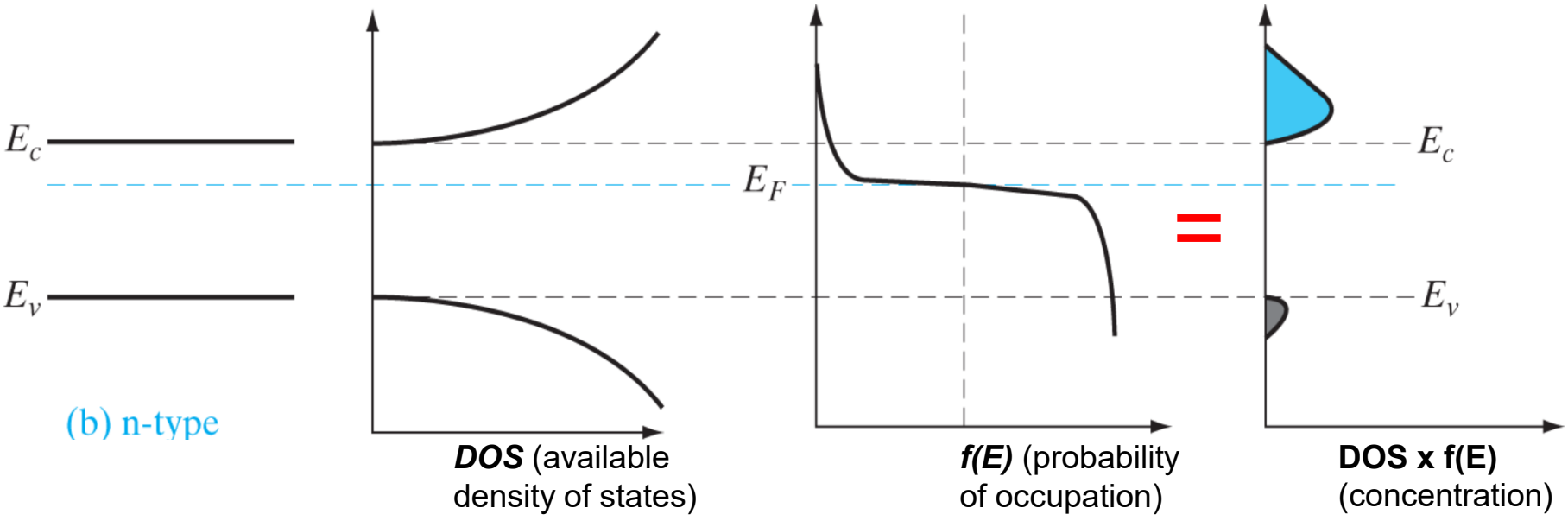
# How Do We Calculate Carrier Densities?

Determining Conductivity

# Answer:

- Integration of [The number (density) of states at a given energy] x [the probability a state is filled at that energy]

$$n_0 = \int_{E_c} f(E) N_c(E) dE$$



# Answer:

- For electrons, we integrate the product of the number (density) of states at a given energy times the probability a state is full at that energy
  - Conduction band only
  - Because the valence band is full, electrons in the valence band do not contribute to the conduction process

$$n_0 = \int_{E_c}^{\square} f(E) N_c(E) dE$$

- For holes, we integrate the product of the number (density) of states at a given energy times the probability a state is empty at that energy
  - Valence band only

$$p_0 = \int_{-\infty}^{E_v} \left[1 - f(E)\right] N_v(E) dE$$

But how do we do that?

# Fermi Distribution and Fermi Level

Determining the Probability of State Occupancy

# Fermions and Bosons

- ***Fermion:*** Only a single fermion can occupy any given quantum state – they are subject to the Pauli exclusion principle. Electrons are fermions, and follow Fermi-Dirac statistics.
- ***Boson:*** There is no limit to the number of bosons that can occupy any given quantum state – they are not subject to the Pauli exclusion principle. Photons are bosons, and follow Bose-Einstein statistics.

# Fermi-Dirac Distribution

- The probability that an available energy state at E will be occupied by an electron at temperature T (temperature in Kelvin) is given by the **Fermi-Dirac Distribution Function**
- Fermi-Dirac distribution function:

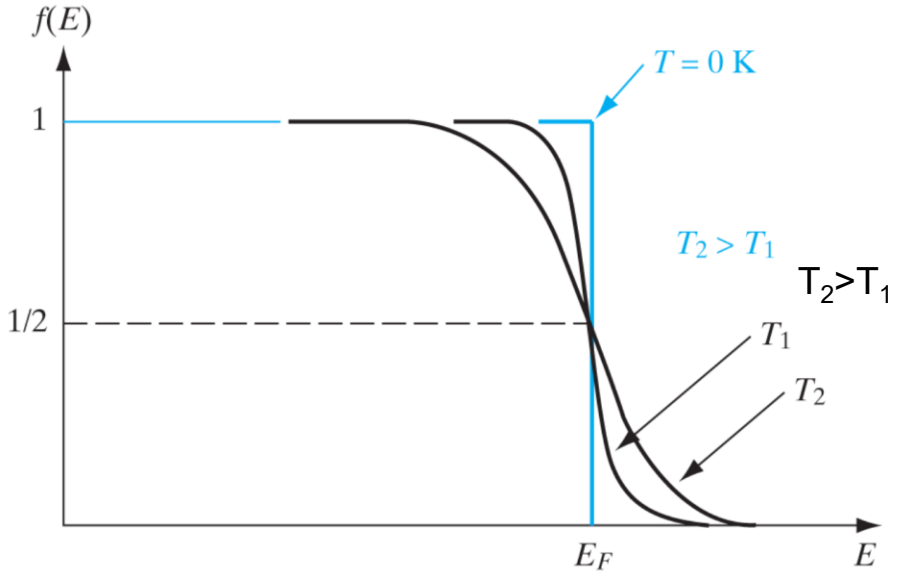
$$f(E) = \frac{1}{1 + e^{(E - E_f)/kT}}$$

$E_f$ : Fermi Level

$k$ : Boltzmann Constant

$$k = 8.62 x 10^{-5}\, eV / K$$

- **Fermi level ($E_f$)**: the energy that determines how likely an electronic states is to be occupied by an electron.
- The Fermi level has units of energy (typically eV)




(a) $E = E_f$ $\quad f(E = E_f) = \frac{1}{1 + e^0} = \frac{1}{2}$

(b) T = 0K

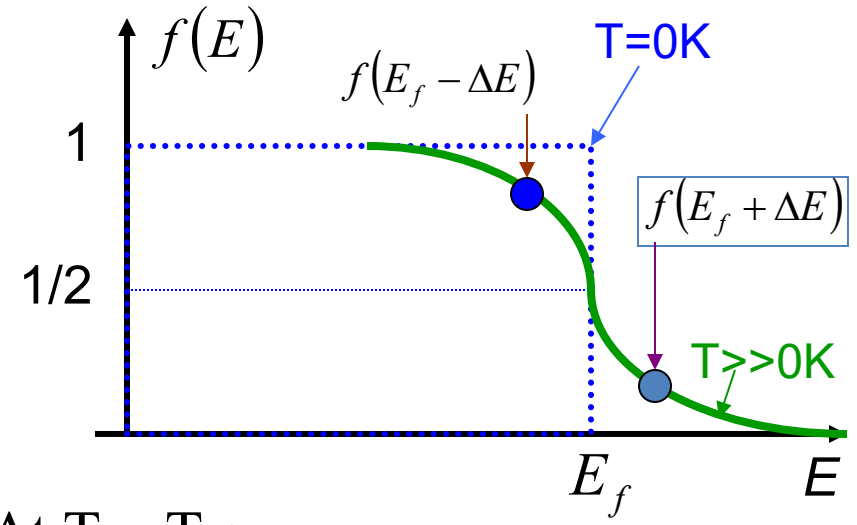
$$f(E < E_f) = \frac{1}{1 + e^{-\square}} = 1$$

$$f(E > E_f) = \frac{1}{1 + e^{+\square}} = 0$$

# What About Holes?

**Hole Probability:**

- A state is either empty or full, so the probability a state is empty is 1-P(full)
- P(full) is given by the Fermi distribution *f(E)*
- P(empty) = 1 - *f(E)*
- $f(E_f+\Delta E)$ is the probability of finding an electron at E+ΔE
- 1-$f(E_f-\Delta E)$ is the probability of finding a hole at E-ΔE




At $T = T_1$ :

$$f(E) = \frac{1}{1+e^{(E-E_f)/kT_1}}$$

$$1-f(E) = \frac{e^{(E-E_f)/kT_1}}{1+e^{(E-E_f)/kT_1}} = \frac{1}{1+e^{(E_f-E)/kT_1}}$$